

SSUI / Stationeers UI Redesign Proposal

Radial Interaction, Smart Storage, and Modern Visor HUD

FlorpyDorp / SSUI discussion notes

Draft v0.1

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Figure 1: Concept direction: streamlined visor HUD with only two hand boxes at the bottom, not a conventional ten-slot hotbar.

1 Executive Summary

This proposal outlines a modernized Stationeers UI redesign that preserves the core identity of the game: **two hands, physical inventory, tools with slots, and engineering complexity**.

The current Stationeers UI exposes a lot of information through inventory boxes and modifier-based selection. While functional, it can feel visually dense, slow during high-pressure gameplay, and less intuitive when dealing with nested bags, batteries, cartridges, and tool-specific actions.

The proposed redesign introduces:

1. **A toolbag radial menu** opened by holding middle mouse.
2. **A tool-specific radial menu** opened by pressing R while holding a tool.
3. **Slot and control branching radials** for tools that have batteries, cartridges, lenses, chips, or other swappable components.
4. **Smart stow behavior** using G, with item-type prioritization and bag-aware stacking.
5. **Custom bag profiles** that define what categories or items belong in each bag.
6. **Nested bag navigation through radials**, preserving the two-hands system while making inventory access faster.
7. **A full visual UI redesign** with a cleaner, more modern HUD that can optionally curve around the screen like a helmet visor.
8. **Optional hardcore visibility rules**, where advanced HUD elements only appear when wearing a helmet, sensor lenses, or equivalent equipment.

The goal is not to simplify Stationeers into a generic survival game. The goal is to make its complexity **faster to read, easier to operate, and more physically coherent**.

2 Background

Stationeers has a deep physical inventory system. Players do not simply carry a traditional RPG inventory. They manage suits, hands, bags, nested containers, cartridges, batteries, tools, filters, tanks, and devices.

That complexity is part of the game's identity.

However, the current UI has several pain points:

- Inventory management depends heavily on box grids.
- Players often need to hold mouse modifier buttons to select items.
- Tool, battery, and cartridge swapping can be clunky.
- Nested bags become difficult to manage as a base grows.
- Critical interactions can require too much screen scanning.
- The UI can become visually busy, especially during construction or emergencies.
- The two-hand system is important, but the interface around it could be more elegant.

From the design conversation, the central idea became:

Radials, radials, radials - but only where they support the physical two-hand logic of Stationeers.

This means the redesign should not become a conventional hotbar system. Stationeers should still feel like a game where you are physically holding tools and managing equipment, not magically accessing ten numbered slots.

3 Core Design Goals

3.1 Preserve the Two-Hands System

The redesign must maintain the vanilla Stationeers logic of:

- Left hand
- Right hand
- Items physically occupying hands
- Tools being equipped into hands
- Bags being opened and navigated intentionally

There should not be a ten-slot conventional hotbar. The bottom UI should show only the two hands.

3.2 Reduce Visual Clutter

The UI should become more streamlined and modern. Instead of large permanent windows covering the screen, information should appear contextually.

The default HUD should focus on:

- Suit and environment essentials
- Two hand slots
- A minimal interaction prompt
- Optional compact vitals
- Radials only when actively invoked

3.3 Make Tool Interaction Faster

Tools often have multiple secondary interactions:

- Turn on/off
- Change mode

- Swap battery
- Swap cartridge
- Swap sensor lens
- Change chip
- Access internal slots
- Perform tool-specific actions

These interactions should be consolidated into a tool radial opened with R.

3.4 Make Bags More Intelligent

Bags should support better organization without removing the physical inventory gameplay.

The proposal adds:

- Smart stow
- Bag profiles
- Category targeting
- Item-specific priorities
- Shareable XML profile files

3.5 Support Both Immersive and Practical UI Modes

Players should be able to choose between:

- Flat modern HUD
- Curved helmet visor HUD
- Hardcore mode where certain UI elements require helmet/sensor equipment

4 Existing UI Problem Areas

Area	Current Issue	Proposed Direction
Tool selection	Requires inventory box interaction or modifiers.	Toolbag radial opened by holding middle mouse.
Tool actions	Secondary actions are scattered or unclear.	R opens contextual tool radial.
Batteries/cartridges	Swapping is awkward.	Slot radial shows compatible replacements.
Nested bags	Hard to browse quickly.	Radial navigation from bag to bag.
Item stowing	Manual and repetitive.	G smart stows based on stack, item type, and bag profile.
Bag organization	Player-created organization is fragile.	Profiles per bag, editable and shareable.
HUD	Busy and box-heavy.	Streamlined visor-style HUD.
Hotbar	Must not become generic 10-slot bar.	Preserve two visible hand boxes only.

5 Proposed Interaction Model

5.1 Two-Hand HUD

The bottom of the screen should show only two primary equipment boxes:

[LEFT HAND]	[RIGHT HAND]
Welder	Wrench

Each hand slot should show:

- Item icon
- Item name
- Durability or charge if relevant
- State indicator if relevant, such as on/off, mode, battery low
- Optional small modifier hint

There should be no numbered ten-slot hotbar.

This preserves Stationeers' unique physical interaction language.

5.2 Toolbag Radial

5.2.1 Input

Hold:

Middle Mouse Button

5.2.2 Behavior

Holding middle mouse opens the **Toolbag Radial**.

This radial shows tools currently available in the player's toolbag.

Example radial entries:

- Welder
- Wrench
- Crowbar
- Drill
- Angle Grinder
- Tablet
- Pipe Bender
- Cable Tool
- Mining Drill
- Screwdriver

Selecting a tool places it into the appropriate hand, using vanilla-like behavior.

5.2.3 Design Notes

The toolbag radial should feel like a faster way of doing what the player already does manually. It should not teleport tools from impossible places or violate inventory logic.

Potential rules:

- Prefer tools in the active toolbelt/toolbag.
- If duplicate tools exist, prioritize the most accessible one.
- If a hand is empty, place the selected tool there.
- If both hands are occupied, use existing Stationeers swap/drop behavior or prompt the player.
- Allow settings for preferred hand behavior.

6 Tool-Specific Radials

6.1 Tool Radial Opened With R

When the player has a tool in hand, pressing:

R

opens a radial specific to that tool.

This radial contains the tool's available actions.

Example: Welder

Welder Radial

- Toggle On / Off
- Adjust Mode
- Slots
- Controls

Example: Tablet

Tablet Radial

- Open Interface
- Change Cartridge
- Change Mode
- Slots

Example: Heavy Miner

Heavy Miner Radial

- Toggle On / Off
- Mining Mode
- Battery Slot
- Filter Slot
- Controls

6.2 Branching Radial for Tools With Slots

For tools that have both actions and swappable components, the first radial level should split into two main branches:

[SLOTS] [CONTROLS]

6.2.1 Slots

The **Slots** branch contains swappable internal components, such as:

- Battery
- Cartridge
- Sensor lens
- Chip
- Filter
- Canister
- Magazine-style component, if applicable

6.2.2 Controls

The **Controls** branch contains active tool commands, such as:

- Turn on/off
- Change mode

- Increase/decrease setting
- Toggle light
- Activate scan
- Eject cartridge
- Configure behavior

This keeps the radial from becoming overloaded.

7 Battery, Cartridge, and Slot Swapping

7.1 Compatible Item Discovery

When the player selects a slot, the radial should show all compatible items available from the player's accessible inventory.

For example, selecting a battery slot could show:

Battery Slot

- Battery: Loose, Backpack
- Battery: Toolbag
- Battery: inside Heavy Miner
- Battery: inside Tablet
- Battery: inside Spare Welder

The system should be able to identify compatible items even if they are inside other tools.

For example:

Battery: inside Heavy Miner

This allows the player to swap batteries without needing to manually open multiple containers, find the tool, remove the battery, then install it into another tool.

7.2 Physical Logic

The system should still respect Stationeers' physical inventory rules.

Possible constraints:

- Only accessible bags/tools are eligible.
- Locked or inaccessible containers are excluded.
- Items inside sealed or unreachable containers are not shown.
- Swapping should update both source and destination items correctly.
- If removing a battery disables another tool, that should be clearly shown.

7.3 Radial Labeling

Each option should clearly state:

[Item Type]: [Location]

Examples:

Battery: Backpack

Battery: Toolbag

Battery: inside Heavy Miner

Cartridge: Tablet

Sensor Lens: Utility Pouch

If a swap would create a consequence, show it subtly:

Battery: inside Heavy Miner

Heavy Miner will be unpowered

8 Smart Stow System

8.1 Input

Press:

G

8.1.1 Behavior

G performs **Smart Stow**.

Instead of simply putting the item into any available space, Smart Stow uses rules to decide the best destination.

8.2 Smart Stow Priority Rules

When the player smart-stows an item, the system should attempt the following:

1. **Stack with existing items of the same type.** Example: steel sheets go to the bag already containing steel sheets.
2. **Prioritize a bag that already contains that item type.** If a bag has steel sheets, future steel sheets should prefer that bag.
3. **Respect explicit bag profiles.** If the player has assigned construction materials to a specific bag, the item should go there.
4. **Respect category rules.** Example: cables, pipes, sheets, and frames go into a construction bag.
5. **Use fallback storage if no matching profile exists.** If no better destination exists, place the item in the default backpack or active bag.
6. **Fail gracefully.** If no valid storage exists, provide a clear message or sound cue.

8.3 Editable Smart Stow Settings

Players should be able to customize Smart Stow behavior.

Setting	Description
Prefer existing stacks	Prioritize bags with matching item stacks.
Prefer item profile	Follow item-specific bag rules first.
Prefer category profile	Follow category-level rules first.
Allow nested bag stow	Permit smart stow into bags inside bags.
Allow tool slot source	Allow smart stow to interact with internal tool slots.
Fallback bag	Choose where unmatched items go.
Full bag behavior	Choose whether to overflow, reject, or prompt.

9 Bag Profiles

9.1 Purpose

Bag profiles allow players to define what belongs in each bag.

Example profiles:

Construction Bag

- Frames
- Sheets

- Cables
- Pipes
- Glass
- Iron
- Steel

Atmospherics Bag

- Pipes
- Pipe valves
- Passive vents
- Active vents
- Filters
- Tanks

Electrical Bag

- Cables
- Cable coils
- APC parts
- Batteries
- Solar parts

Farming Bag

- Seeds
- Fertilizer
- Plant trays
- Bottles
- Organics

9.2 Per-Bag Dropdown

Each bag should have a small profile dropdown.

Example:

Bag Profile: Construction

Changing the dropdown assigns that bag's organizational role.

This lets players carry multiple bags with clear purposes.

9.3 XML-Based Sharing

Profiles should be stored in an XML-based format so players can:

- Copy profiles between saves
- Share profiles with other players
- Build server-standard loadouts
- Publish recommended bag organization setups
- Import/export community profiles

Example concept:

```
<BagProfile name="Construction">
  <Category name="BuildingMaterials" priority="100" />
  <Item prefab="ItemSteelSheets" priority="100" />
  <Item prefab="ItemIronFrames" priority="90" />
  <Item prefab="ItemCableCoil" priority="80" />
</BagProfile>
```

This does not need to be the final schema. It is only a conceptual example.

10 Bag Radial Navigation

10.1 Backpack Radial

Opening the backpack should present a radial view of available nested bags or major storage sections.

Example:

Backpack Radial

- Construction Bag
- Atmospherics Bag
- Electrical Bag
- Farming Bag
- Loose Items
- Tools

Selecting a bag opens that bag's radial.

10.2 Nested Bag Radials

If the player selects the Construction Bag:

Construction Bag

- Sheets
- Frames
- Cables
- Pipes
- Glass
- Open Full Inventory

Selecting a category can open another radial or compact list.

The goal is to make nested bags usable without turning the interface into a giant box grid.

10.3 Item Selection

Selecting an item from a bag radial should behave like vanilla inventory interaction:

- Put item in empty hand if available.
- Swap with held item if appropriate.
- If both hands are full, prompt or use the configured behavior.
- Preserve the physical logic of items being held and moved.

11 Visual UI Redesign

11.1 Overall Direction

The UI should feel more modern, readable, and cinematic.

Visual goals:

- Less box clutter
- More contextual information
- Better spacing
- Higher contrast
- Clearer hierarchy
- Subtle sci-fi styling
- Better support for dark environments
- Optional visor curvature

The redesign should take inspiration from modern interfaces like Satisfactory or Helldivers-style readability, while remaining distinctly Stationeers.

11.2 Curved Visor Mode

The player should be able to enable a HUD style where UI elements appear to curve around the inside of a helmet visor.

This mode would include:

- Curved top status band
- Curved lower hand UI
- Subtle side visor lines
- Minimal edge indicators
- Slight translucent glass effect
- Optional distortion or parallax

This should be a visual option, not a forced gameplay change.

11.3 Flat HUD Mode

Some players may prefer a conventional flat HUD.

The redesign should allow:

HUD Curvature: Off / Low / Medium / High

This keeps the mod usable for players who dislike helmet-style curved UI.

11.4 Hardcore HUD Mode

A hardcore mode could restrict advanced HUD elements unless the player has appropriate equipment.

Equipment	HUD Elements Enabled
No helmet	Minimal or no advanced HUD.
Suit helmet	Basic pressure, O2, temperature, vitals.
Sensor lenses	Enhanced scanning, device overlays.
Tablet/cartridge	Device/network information.
Advanced suit	Full visor interface.

This could make the UI feel more diegetic and survival-oriented.

12 Proposed Default HUD Layout

12.1 Top Status Strip

A slim curved top band should show only essentials:

Pressure | O2 | Temperature | Power | Water | Suit Status

This should replace large always-open panels.

12.2 Center Reticle

A small contextual reticle remains in the center.

When looking at an object, a small interaction panel appears.

Example:

Fabricator
Status: Online
Queue: 1/2
[F] Configure
[E] Pick Up

The center panel should be compact and disappear when not needed.

12.3 Bottom Two-Hand UI

The bottom center shows only:

Left Hand	Right Hand
Welder	Wrench

Each hand box may show:

- Icon
- Item name
- Charge
- Mode
- Warning state
- Held quantity, if stackable

No numbered hotbar.

12.4 Compact Vitals Panel

A small vitals panel can sit in a corner.

Example:

Vitals
Health: 100%
O2: 68%
Power: 78%
Temp: 21 deg C

This should be readable but not dominant.

12.5 Inventory Prompt

Inventory access can be represented by a small prompt:

[TAB] Inventory

The full inventory should not always occupy the screen.

13 Controls Summary

Input	Action
Middle Mouse Hold	Open toolbag radial.
Release Middle Mouse	Confirm selection / close radial.
R while holding tool	Open tool-specific radial.
G	Smart stow held item.
TAB	Open inventory / backpack radial.
F	Configure targeted device.
E	Pick up / interact with targeted device.

These are draft bindings and should be configurable.

14 Example User Flows

14.1 Selecting a Tool

1. Player holds middle mouse.
2. Toolbag radial opens.
3. Player highlights Wrench.
4. Player releases middle mouse.
5. Wrench is placed in hand.

Result: Same core action as vanilla, but faster and cleaner.

14.2 Swapping a Welder Battery

1. Player is holding Welder.
2. Player presses R.
3. Welder radial opens.
4. Player chooses Slots.
5. Player chooses Battery.
6. Radial shows compatible batteries:
 - Battery: Backpack
 - Battery: Toolbag
 - Battery: inside Heavy Miner
7. Player selects battery.
8. Battery is swapped into Welder.

Result: The player does not need to manually dig through multiple inventories.

14.3 Smart Stowing Steel Sheets

1. Player holds steel sheets.
2. Player presses G.
3. System checks for existing steel sheet stacks.
4. Finds steel sheets in Construction Bag.
5. Adds sheets to that stack.

Result: Construction materials remain organized automatically.

14.4 Using a Bag Profile

1. Player opens backpack.
2. Selects a nested bag.
3. Opens profile dropdown.
4. Assigns Atmospherics.
5. Future pipes, valves, vents, filters, and tanks prioritize that bag.

Result: Bag organization becomes predictable and shareable.

15 Multiplayer Considerations

Because Stationeers multiplayer can be sensitive to mod behavior, the redesign should be careful about client/server logic.

Potential concerns:

- Inventory state synchronization
- Tool slot swapping
- Nested bag updates
- Race conditions when moving items quickly
- Multiplayer desync when swapping batteries from inside another tool
- Server authority over item movement
- Compatibility with vanilla saves
- Compatibility with existing mods

The UI can be client-side, but item movements must respect authoritative game logic.

16 Potential Development Phases

16.1 Phase 1: Visual HUD Prototype

- Two-hand HUD only
- Curved visor overlay option
- Minimal top status strip
- Context interaction panel
- No functional inventory changes yet

16.2 Phase 2: Toolbag Radial

- Middle mouse radial
- Select tool from toolbag
- Equip into hand
- Preserve vanilla behavior

16.3 Phase 3: Tool-Specific Radial

- R opens tool menu
- Controls branch
- Slots branch
- Simple battery swap support

16.4 Phase 4: Smart Stow

- G smart stow
- Stack prioritization
- Item-type destination memory
- Settings menu support

16.5 Phase 5: Bag Profiles

- Per-bag dropdown
- Category and item rules
- XML import/export
- Shareable profiles

16.6 Phase 6: Nested Bag Radials

- Backpack radial
- Bag-to-bag radial navigation
- Category views
- Fallback full inventory option

16.7 Phase 7: Hardcore / Diegetic HUD Mode

- Helmet-dependent HUD
- Sensor-lens-dependent overlays
- Optional advanced immersion settings

17 Open Questions

17.1 Interaction

- Should tool radial open on press R or hold R?
- Should middle mouse radial confirm on release or click?
- How should the system decide left hand vs right hand?
- What happens if both hands are occupied?
- Should the player be able to assign preferred tools to preferred hands?

17.2 Inventory Logic

- How deep should smart stow search nested bags?
- Should smart stow be allowed to move items into closed nested bags?
- Should batteries inside other tools always be available for swapping?
- Should the system warn before pulling a battery from an important tool?

17.3 Bag Profiles

- Should profiles be per save, global, or both?
- Should servers be able to enforce default profiles?
- Should profiles support modded items automatically?
- Should unknown modded items fall back to category inference?

17.4 Visual UI

- Should curved visor mode be default or optional?
- Should the HUD curvature be adjustable?
- Should hardcore mode be a gameplay setting, server setting, or client preference?
- How much information should be hidden without a helmet?

17.5 Official Handoff

- If this becomes strong enough, should it be shown to Simon, Dean, or the RocketWerkz team?
- If offered officially, what UI assets need to be remade without AI-generated elements?
- Should this remain a mod, become a proposal, or both?

18 Design Philosophy

The purpose of this redesign is not to remove friction from Stationeers entirely.

Stationeers is supposed to be physical, dangerous, and demanding. The player should still feel like they are managing real tools, real bags, real gases, real power, and real consequences.

The current proposal tries to distinguish between **good friction** and **bad friction**.

18.1 Good Friction

- Having only two hands
- Choosing what tools to carry
- Managing batteries and cartridges
- Organizing bags
- Being vulnerable without proper equipment
- Needing to think under pressure

18.2 Bad Friction

- Fighting with inventory boxes
- Losing track of where items are
- Manually digging through nested bags for one battery
- Repeating the same stow action dozens of times
- Having important tool actions hidden behind awkward UI
- Covering the screen with too many panels

The redesign should reduce bad friction while preserving good friction.

19 Draft One-Sentence Pitch

SSUI redesigns Stationeers' inventory and HUD around modern radial interactions, smarter bag organization, and a cleaner visor-style two-hand interface while preserving the game's physical inventory logic and survival-engineering complexity.

20 Immediate Next Steps

1. Refine the two-hand HUD mockup.
2. Define exact radial menu behavior.
3. Write a complete tool interaction spec.
4. Draft XML structure for bag profiles.
5. Identify which systems are UI-only and which require gameplay logic.
6. Create a small MVP scope for Jackson's SSUI update.
7. Decide whether this is being framed as a mod, a pitch to RocketWerkz, or both.